PHASE 1 OF 2

Predicting Tox21 toxicity from molecular structure

Morgan fingerprints, RDKit descriptors and classical machine learning

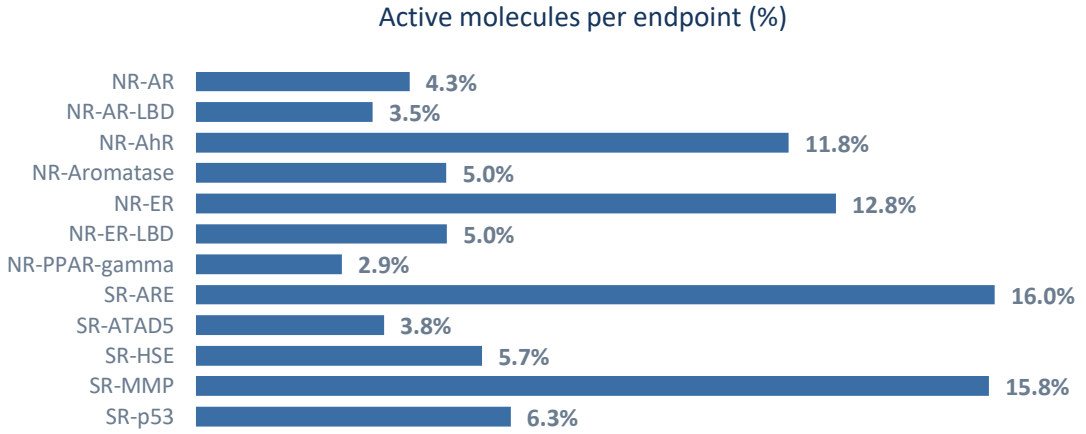
Logistic Regression · Random Forest · XGBoost

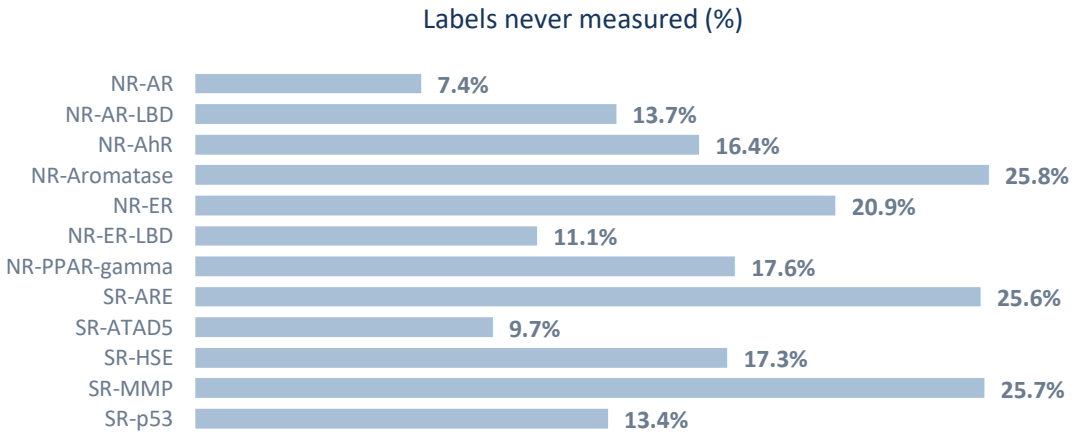
Avneet Singh | Khushi | Manpreet Singh | Riya Yadav | Satvik Prasad

B.Tech Computer Science and Artificial Intelligence · 7,708 molecules, 12 assay endpoints, one classifier per endpoint

Problem and dataset

- 7,708 unique molecules after cleaning, each given as a SMILES string
- 12 toxicity endpoints — 7 nuclear receptor (NR-*) and 5 stress response (SR-*) — one binary classifier per endpoint
- 17.05% of all label cells are missing: not tested, not non-toxic. Missing labels stay NaN and every metric uses labelled rows only
- Labelled molecules per endpoint: 5,716 to 7,141
- Positive rate per endpoint: 2.91% (NR-PPAR-gamma) to 15.95% (SR-ARE)





WHY ACCURACY IS NOT REPORTED

Roughly one molecule in ten is active, so a model answering “inactive” for everything already scores about 89.5% accuracy. We report ROC AUC and PR AUC instead; a random model's PR AUC equals the positive rate.

The question the project exists to answer

A molecule can be shown to a model in two ways. Which one works better is contested, so we built the classical side properly before adding a graph model.

A Fixed representation

Designed in advance — never changes

- Morgan fingerprint: 2,048 bits of local substructure
- 12 RDKit descriptors: weight, LogP, TPSA, rings
- Fed to a classical classifier

PHASE 1 COMPLETE

B Learned representation

Learned during training — adapts to the task

- The molecule stays a graph: atoms as nodes, bonds as edges
- A graph neural network builds its own features
- Nothing about the representation is fixed in advance

PHASE 2 NOT YET RUN

RESEARCH QUESTION

Do learned molecular graph representations outperform fixed fingerprint-based representations, when both are evaluated on the same data split under the same protocol?

Pipeline, and who owns each stage



Seed 42 fixed at every stage

Missing labels masked never filled with 0

Test split read once after every choice was frozen

Cleaning, and a scaffold split rather than a random one

LED BY KHUSHI AND RIYA YADAV

CLEANING FUNNEL

7,831 **Raw Tox21 file**

removed 8 **SMILES RDKit cannot sanitise**

All eight are aluminium compounds writing [AlH3] with three extra bonds

244 **Salts and mixtures reduced to largest fragment**

Then written as canonical SMILES

removed 115 **Duplicate rows merged**

187 rows collapse to 72 molecules; 47 conflicting labels set to missing

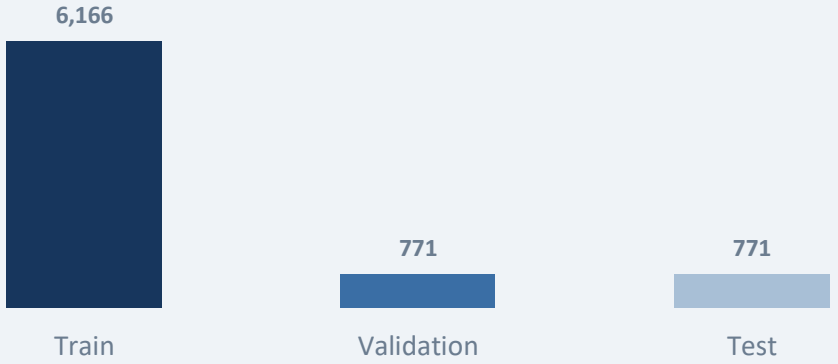
7,708 **Clean dataset**

Unique canonical SMILES

Merging is not tidying. Without it the same molecule sits in train and test under two salt forms, and every reported score is inflated.

BEMIS–MURCKO SCAFFOLD SPLIT

Molecules are grouped by ring scaffold and whole groups are assigned to one partition, so every evaluation molecule carries a ring system the models never saw. Verified: no molecule, canonical SMILES or scaffold is shared across partitions.



80 / 10 / 10 · 2,285 unique scaffolds

Known limitation. Acyclic molecules have no ring scaffold, so all 1,753 form one group and land in training. Validation and test hold ring systems only.

Features: Morgan fingerprints and RDKit descriptors

LED BY MANPREET SINGH

morgan	descriptors	morgan+descriptors
2,048 bits	12 columns	2,060 columns
radius 2 (ECFP4 setting)	RDKit 2D physicochemical properties	2,048 bits + 12 descriptors

HOW A MORGAN FINGERPRINT IS BUILT

1 Every heavy atom gets an identifier from its element, degree, formal charge, attached hydrogens and ring membership.

2 Twice over, each identifier is recomputed from itself plus its neighbours' bond types and identifiers.

3 After two rounds an identifier summarises the substructure within two bonds of that atom.

4 Each identifier is folded into the vector by id mod 2048, and that bit is set to one.

WHAT THE 12 DESCRIPTORS COVER

- Size and bulk: molecular weight, heavy atoms, heteroatoms
- Lipophilicity: MolLogP
- Polarity and hydrogen bonding: TPSA, donors, acceptors
- Shape and flexibility: rotatable bonds, rings, aromatic rings, fraction of sp^3 carbons, formal charge

HASH COLLISIONS ARE AN ACCEPTED COST

Far more substructures exist than bit positions, so two environments can map to the same bit. A set bit means “at least one substructure hashing here is present”, not that a specific fragment is present. This is the price of a compact, fixed-length vector.

NO LEAKAGE INTO EVALUATION

Median imputation and standardisation are fitted on the training partition only. Fingerprint bits pass through unscaled; only the descriptor columns are imputed and standardised.

Models, tuning and evaluation protocol

LED BY AVNEET SINGH AND SATVIK PRASAD

Model	Family	Searched on validation ROC AUC
Logistic Regression	Linear	C ∈ {0.001, 0.01, 0.1, 1, 10}; class_weight ∈ {none, balanced}
Random Forest	Bagging	300 trees, max_features = sqrt; min_samples_leaf ∈ {1, 3}; class_weight ∈ {none, balanced_subsample}
XGBoost	Boosting	max_depth ∈ {3, 6}; learning_rate ∈ {0.05, 0.1}; n_estimators ∈ {100, 300}; scale_pos_weight ∈ {1, balanced}

Hyperparameters are chosen per endpoint, not once per model — 12 independent searches for each model and feature set.

108

classifiers trained

3 models × 3 feature sets × 12 endpoints, each with its own tuned configuration

PROTOCOL

- One binary classifier per endpoint, trained on that endpoint's labelled training rows only
- Class imbalance is a tuned hyperparameter, not an assumption: class_weight (LR), balanced_subsample (RF), scale_pos_weight (XGB)
- Selection metric: mean validation ROC AUC, with average precision as the tie-breaker
- The test split is scored once, after every hyperparameter choice is fixed

WHY THESE TWO METRICS

ROC AUC is the probability that a random active molecule is ranked above a random inactive one. It is unaffected by how rare positives are.

PR AUC summarises the precision–recall curve and is far more sensitive to the rare positive class.

Results: scaffold test split

LED BY SATVIK PRASAD

Model	Features	Val ROC	Val PR	Test ROC	Test PR
Logistic Regression	morgan	0.7239	0.3230	0.7016	0.2774
Logistic Regression	descriptors	0.7232	0.2216	0.6662	0.2188
Logistic Regression	morgan+desc	0.7486	0.2970	0.7386	0.2688
Random Forest	morgan	0.7497	0.3825	0.7164	0.3173
Random Forest	descriptors	0.7433	0.3325	0.7361	0.3010
Random Forest	**morgan+desc**	**0.7850**	**0.4069**	**0.7635**	**0.3702**
XGBoost	morgan	0.7366	0.3495	0.7125	0.2815
XGBoost	descriptors	0.7490	0.3368	0.7104	0.2686
XGBoost	morgan+desc	0.7739	0.3906	0.7552	0.3541

Mean over the 12 endpoints, single scaffold split, seed 42. Bold = best row. No confidence intervals: one split and one seed were run.

0.7635

Best test ROC AUC

Random Forest, morgan+desc

0.3702

Best test PR AUC

Same model — it leads on both

–0.026

Validation → test drop

Mean ROC AUC across the nine configurations

READING THE TABLE

- Random Forest on morgan+desc is strongest on every column: validation ROC 0.7850 and test ROC 0.7635, test PR 0.3702
- XGBoost on morgan+desc is close behind at 0.7552 / 0.3541; the gap to Random Forest is about 0.008 ROC, which one split cannot resolve

Do descriptors help? Paired gain from adding them

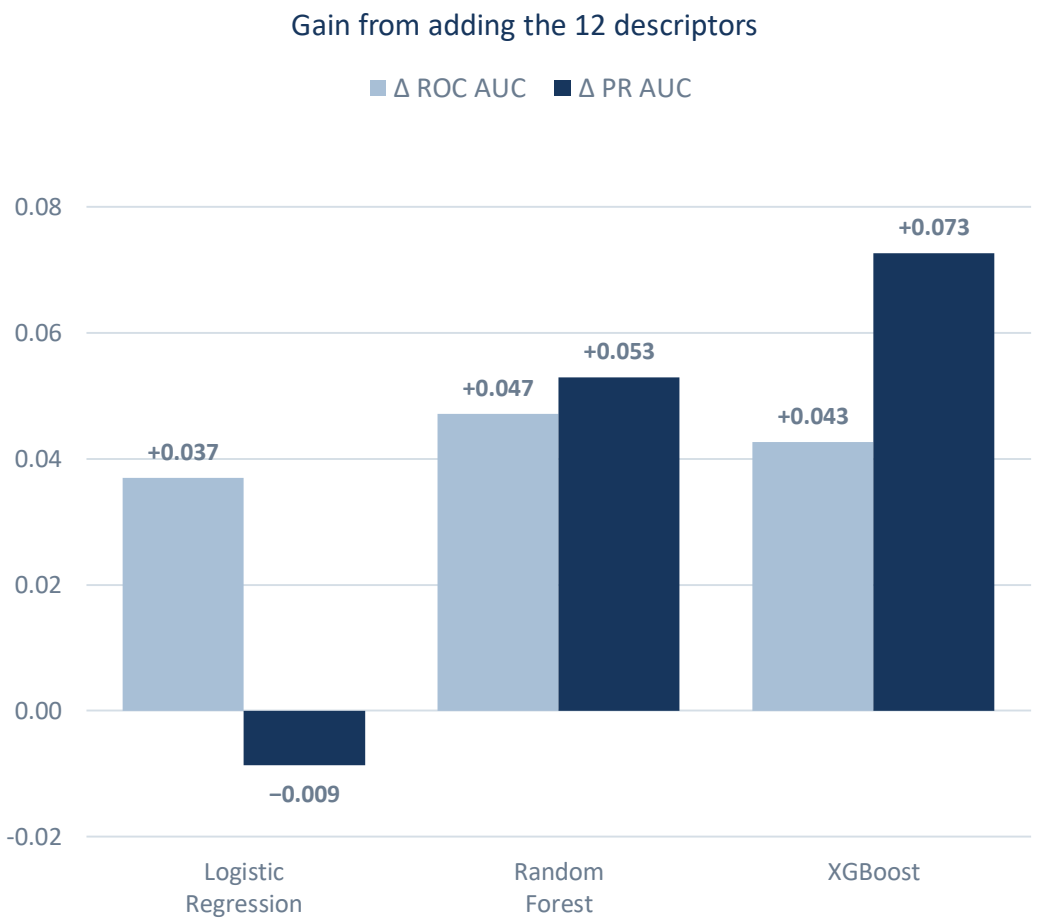
LED BY SATVIK PRASAD

Model	Δ ROC AUC	Δ PR AUC
Logistic Regression	+0.0370	**−0.0086**
Random Forest	+0.0471	+0.0529
XGBoost	+0.0427	+0.0726

Δ = morgan+descriptors minus morgan, scaffold test split, same endpoints and same split.

WHAT THIS SHOWS

- All three models gain ROC AUC from the descriptors — between +0.037 and +0.047
- On PR AUC the picture splits. The tree models gain substantially (+0.053 and +0.073), while Logistic Regression actually loses a little (−0.009)
- A linear model cannot combine 2,048 sparse bits with 12 dense descriptors the way a tree can; the trees exploit interactions between the two kinds of feature
- We did not run an ablation, so we report the effect without claiming the mechanism



Per-endpoint results (scaffold test, morgan+descriptors)

LED BY SATVIK PRASAD

Endpoint	Pos.	LR ROC	RF ROC	XGB ROC	LR PR	RF PR	XGB PR
NR-AR	14	0.539	0.496	**0.574**	0.023	0.024	**0.027**
NR-AR-LBD	12	0.780	**0.795**	0.762	0.104	**0.231**	0.217
NR-AhR	110	0.814	0.829	**0.833**	0.529	0.596	**0.613**
NR-Aromatase	53	0.811	**0.826**	0.798	0.344	**0.518**	0.452
NR-ER	72	0.671	**0.738**	0.714	0.234	**0.395**	0.385
NR-ER-LBD	24	0.676	**0.717**	0.693	0.165	0.212	**0.214**
NR-PPAR-gamma	32	0.741	**0.796**	0.749	0.198	0.328	**0.342**
SR-ARE	120	0.744	**0.803**	0.786	0.466	**0.611**	0.568
SR-ATAD5	33	0.809	**0.826**	0.798	0.152	**0.295**	0.208
SR-HSE	55	0.738	0.750	**0.767**	0.298	0.348	**0.379**
SR-MMP	89	0.814	0.807	**0.819**	0.457	**0.494**	0.479
SR-p53	75	0.727	**0.779**	0.769	0.255	**0.391**	0.368

Bold = highest in that row for that metric. “Pos.” is the number of active molecules in the test split for that endpoint.

RANDOM FOREST LEADS, BUT NOT EVERYWHERE

Highest ROC AUC on 8 of 12 endpoints and highest PR AUC on 7; XGBoost takes the other 4 and 5.

LOGISTIC REGRESSION LEADS ON NONE

With morgan alone it led on three endpoints. Once descriptors are added, it leads on none — the ensembles use them better.

TWO ENDPOINTS ARE TOO SMALL TO READ

NR-AR has 14 test actives and NR-AR-LBD has 12. Their scores swing on a handful of molecules and we do not over-interpret them.

BEST AND WORST

Strongest: NR-AhR 0.833 and SR-MMP 0.819. Weakest: NR-AR, where no model clears 0.58.

The trained models, running in a browser

LED BY MANPREET SINGH

WHY THIS MATTERS FOR THE PROJECT

A model nobody can run is hard to examine. All three classical models were exported and now run entirely in the browser — no server, no hosting cost, and the user's molecule never leaves their machine.

1. **RDKit compiled to WebAssembly**
   Reproduces the exact cleaning and fingerprint steps from the notebooks, in the browser.

2. **Trees serialised to a compact binary format**
   Evaluated directly in JavaScript, so there is no Python runtime involved.

3. **Any SMILES string, all 12 endpoints**
   The user types a molecule and gets 12 probabilities from each of the three models.

`toxilab.man9876tr.workers.dev`

CORRECTNESS IS CHECKED, NOT ASSUMED

55,512

in-browser predictions compared against the Python models, molecule by molecule

4.2×10^{-7} largest disagreement

REPRODUCIBILITY

- Re-running the preprocessing notebook on the raw Tox21 file reproduces the cleaned dataset, the three split files and the report byte for byte
- Logistic Regression reproduces exactly; Random Forest and XGBoost differ by at most 1×10^{-5} ROC AUC from parallel tree construction
- Repository holds every trained model, all metrics, per-molecule predictions and 20 automated data-integrity tests

Team contributions and Phase 2

Member	Contribution
Khushi	Exploratory data analysis (EDA): dataset inspection, label distribution, missing-value analysis, class imbalance and EDA visualisations.
Riya Yadav	Data-preprocessing support; literature review, research background, report writing, documentation and presentation content.
Manpreet Singh	Morgan fingerprint generation, molecular feature representation and feature-engineering workflow.
Avneet Singh	Major model-training work: Logistic Regression, Random Forest and XGBoost experiments, tuning and model comparisons.
Satvik Prasad	Evaluation workflow, endpoint-wise metrics, results analysis, visualisations and project integration/testing support.

- **Phase 2:** implement a graph neural network (GCN) on the same scaffold split and evaluation protocol, then compare it with the completed classical ML baselines.
- **Current completed results cover Phase 1:** Logistic Regression, Random Forest and XGBoost. GNN results are not yet claimed.

LIMITATIONS AND WHAT COMES NEXT

What this baseline does and does not establish

ESTABLISHED

- A cleaned, leakage-free dataset on one frozen scaffold split shared by every model
- Adding descriptors to fingerprints improves ROC AUC for all three models
- Random Forest on morgan+descriptors is strongest: test ROC 0.7635, PR 0.3702
- 108 tuned classifiers, reproducible from the repository and deployed in the browser

NOT ESTABLISHED

- One split and one seed, no confidence intervals — the RF–XGBoost gap of ~0.008 ROC is not resolved
- NR-AR and NR-AR-LBD have 14 and 12 test actives; those endpoint scores are noise
- All 1,753 acyclic molecules are in training, so we characterise none of them
- No graph results are claimed. Phase 2 has not been run

NEXT: PHASE 2

A Graph Convolutional Network with a masked loss that excludes missing labels, trained on the same frozen scaffold split and scored with the same two metrics.

Because the split, the masking and the metrics are already fixed, the comparison will be fair by construction — which is the whole point of Phase 1.

These models predict activity in specific in-vitro Tox21 assays. They are research outputs, not safety assessments, and must not be used for regulatory, clinical or medical decisions.

Thank you · Questions